

| Question |     |       |  | Marking details                                                                                                                                                                                                           | Marks available |          |          |          |          |          |
|----------|-----|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |       |  |                                                                                                                                                                                                                           | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 6        | (a) | (i)   |  | Substitution: work done = $450 \times 0.8$ (1)<br>$\times 2$ (1)<br>$= 720$ (1) [J]<br>Answer 360 [J] award 2 marks<br><br><b>Alternative:</b><br>$2 \times 0.8 = 1.6$ [m] (1)<br>$450 \times 1.6$ (1)<br>$= 720$ (1) [J] | 1<br>1          | 1        |          | 3        | 3        |          |
|          |     | (ii)  |  | Answer from (i) <b>ecf</b> expect 720 [J] <b>or</b> 720 [J]                                                                                                                                                               | 1               |          |          | 1        |          |          |
|          |     | (iii) |  | [Work done in overcoming] friction / lifting the forks<br><b>or</b> [energy transferred as] sound / heat                                                                                                                  |                 | 1        |          | 1        |          |          |
|          | (b) |       |  | $450 - (12 \times 10) = 330$ [N] (1)<br>$\frac{330}{10} = 33$ (1) [kg]<br><br><b>Alternative:</b><br>$\frac{450}{10} = 45$ (1) [kg]<br>$45 - 12 = 33$ (1) [kg]                                                            |                 | 2        |          | 2        | 2        |          |
|          |     |       |  | <b>Question 6 total</b>                                                                                                                                                                                                   | <b>3</b>        | <b>4</b> | <b>0</b> | <b>7</b> | <b>5</b> | <b>0</b> |